TIDES Phase 11

Performance Optimization, Observability & Release Readiness

Evidence brief | 22 October 2026 | Simulator-only validation

Confirmed Fixes

Fixed a conditional React hook-order defect in UnifiedOperationsCenter .

Fixed drone vision payload keys so POST /api/vision/detections satisfies the FastAPI schema.

Replaced hanging Playwright canvas enumeration with bounded canvas count and first-canvas checks.

Added browser smoke coverage asserting zero HTTP 422 responses.

Measured Baseline

Regression Results

Remaining Limitations

JavaScript remains approximately 3.22 MB minified and triggers the existing chunk-size warning. No route-level lazy-loading boundary was introduced without stronger dependency measurement.

Lint has zero errors but 843 pre-existing warnings, including unused imports, hook dependency warnings, render-time nondeterminism, and ref-access warnings.

Hardware, field telemetry, and real-world detection performance remain unvalidated.